

Aditya Sharma

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PROFESSIONAL SUMMARY

AI Engineer with strong foundations in mathematics, statistics, and systems-level engineering. Experienced in building agentic LLM systems, multimodal pipelines, RAG, model deployment using vLLM/Transformers, and GPU-optimized workloads with **CUDA**. Skilled in Python, PyTorch, LangChain/LangGraph, C++, and fine-tuning LLMs (LoRA/QLoRA). Passionate about scalable AI, inference optimization, and applied AI.

EDUCATION

- **Indian Institute of Information Technology** Lucknow, India
M.S - AI and ML; CGPA: 9.23 Dec 2023 - June 2025
- **Rajasthan University** Jaipur, India
B.S (Honors) Mathematics; CGPA: 8.00 July 2020 - June 2023

EXPERIENCE

- **Datasmith.ai — Associate AI Engineer** Oct 2024 – Present **Pune**
 - **MHADA-Sathi (Govt. of Maharashtra Chatbot)** — Designed and deployed the on-premise, multilingual (Marathi and English) agentic chatbot for MHADA, publicly available at: mhada.gov.in/en.
 - * Built an **8-tool agent** (RAG + structured DB tools) using hybrid search and CUDA-accelerated embeddings, achieving **sub-4s latency** across website data, tenders, NOC, lottery, and e-Office.
 - * Performed GPU capacity planning using **KV-cache**, **TTFT**, and **TPS** metrics; secured approval for **4×A100 80GB**.
 - * Deployed:
 - **Chat Model:** [Qwen3-14B-AWQ](#)
 - **Vision Model:** [Qwen2-VL-7B-Instruct](#)
 - **STT:** [Whisper Large-v3](#)
 - **TTS:** [Indic-Parler-TTS](#)
 - using **vLLM with horizontal scaling** and network-level routing across all GPUs to support **100+ concurrent users** for chatbot.
 - * Managed **Postgres recovery**, deployment multiple services like redis, celery, qdrant, postgres, superset, which were used in multiple use cases like **lease extraction**, **petition summarisation**.
 - **Automation for Savills Asia** — Delivered enterprise automation workflows for Savills Asia (savills.com), **directly reporting to the Asia Head of IT Business Strategy**.
 - * Integrated Microsoft Teams M365 with **Copilot Studio** to enable department-level SharePoint and public-site automation.
 - * Implemented authentication/authorization using **Microsoft Entra ID**, App Registrations, Azure Functions, Logic Apps, and Graph API.
 - * Built automated property-management chatbots that preserved **SharePoint user-level permissions** using copilot-studio.
 - **Virtual Data-Lake PoC** — Built a multi-database virtual query engine using **Knowledge Graphs**.
 - * Created intent-recognition and KG-generation pipelines using **Neo4j**, identifying explicit and implicit inter-database relationships.
 - * Enabled **NL2SQL/NoSQL** queries across MongoDB, Postgres, and Neo4j with schema-aware sequential/parallel execution.
 - * Delivered adaptive visualization (tables, pie charts, bar charts) using dynamic React components.
 - **Tech Stack:** RAG, LangGraph, LangChain, CUDA, vLLM, Transformers, Nginx, Docker, Neo4j, MongoDB, Postgres, Azure Graph API, Entra ID, Whisper, Qwen models.
- **PG-AGI Consultancy — AI/ML Intern** May 2024 – June 2024 **Remote**
 - Developed Natural Language Querying PoC for multi-database querying.
 - Built a Custom GPT model tailored for specific client requirements.

IMPLEMENTED PAPERS

- **Attention Is All You Need**
- **Linformer: Self-Attention with Linear Complexity**
- **Playing Atari with Deep Reinforcement Learning**



PERSONAL PROJECTS

- **Upstox & AlphaVantage MCP Trading Server**  [\[Demo\]](#)
 - Built an MCP-based trading server integrating **Upstox APIs** for real-time portfolio access and seamless Buy/Sell order execution.
 - Implemented automated **stock analysis** workflows using the **AlphaVantage API** with indicators, intraday signals, and analytics.
 - Connected the MCP server through the **VS Code MCP client** using **Copilot Studio Agent Mode**, enabling local execution of trading commands such as portfolio fetch, analysis, and **order placement.**
 - Natural language order execution, **Intraday**, **AMO** etc.
- **Reference Reader** 
 - *Tech:* Python, FastAPI, **LangChain**, **OpenAI**, **PostgreSQL**, **Qdrant**, AWS EC2, Docker
 - Built an **AI research assistant** with a global vector store of **ArXiv papers**.
 - Implemented semantic search using Qdrant + PostgreSQL metadata storage.
 - Integrated SERP + Wikipedia APIs for retrieval enhancement.
 - Added **advanced metadata filtering**, improving search relevance.
- **Advanced Knowledge Management Cybershield** 
 - *Tech:* **Knowledge Graph**, Python, Vercel, **OpenAI API**, **NLP**
 - Built AI-driven system to analyze **CVE vulnerability data**.
 - Used Knowledge Graph for extraction of entities and relationships.
 - Implemented **custom NER models** for identifying security patterns.
 - Designed **threat intelligence mapping** using RAG workflow.
- **Pretraining a Transformer on Poetry** 
 - *Tech:* Python, **PyTorch**, **CUDA**, **Multihead Attention**, **TikToken**, **GELU**
 - Scraped **11K poems** and created a custom dataset.
 - Implemented a character-level transformer from scratch.
 - Used multihead attention, GELU, and custom training loop.
 - Applied **gradient accumulation** to handle 16GB T4 GPU.
 - Explored **mixed precision** and other techniques for optimisation according to the GPU

TECHNICAL SKILLS

- **Programming:** Python , C++, JavaScript(Beginner) and Typescript (Beginner)
- **LLM Technologies:** LangChain and Langgraph, PEFT, Transformers,DSPy,vLLM,Nginx Reverse Proxy , Cuda(beginner), redis, celery, Tooling, multi agent orchestration.
- **Tools & Platforms:** Docker, Docker-compose AWS, Azure Devops, Hugging Face, Git, FastAPI, Kubernetes(Beginner), linux